Title: Influence of Neighborhood Environments Features on Long-Term Changes in Physical Activity Across the Lifespan

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INTRODUCTION

- Physical activity (PA) is essential for preventing and controlling diseases and related mortality across the lifespan.
- Many cross-sectional studies shows positive associations of neighborhood environment features (e.g., walkability) with PA.
- Fewer prospective studies have examined the combinations of macro- and micro-scale environment features on PA, mostly focused on single age groups and included short follow-up durations (1-5 years)
- This study examined how baseline macro and micro activity-supportive neighborhood environment features are associated with long-term (5.1 to 15-year) changes in MVPA across multiple age groups (i.e., children, adolescents, adults, older adults)

METHODS

- Secondary analyses leveraging data from 4 cross-sectional age-specific studies on neighborhood environment and PA in 4 age groups (children, adolescents, adults, and older adults).
- 5th study re-recruited participants from previous 4 studies for long-term (5.1-15 yrs follow-up) assessments for 1067 sample
- All studies used a 2 (high/low walkability) x 2 (high/low income) sampling design
- MVPA measured with Actigraph accelerometer at baseline and follow-up
- Perceived neighborhood environment measured with NEWS
- Longitudinal structural path models was used to predict MVPA at follow-up from baseline NEWS scores, adjusting for baseline outcomes and covariates

Baseline activity supportive-neighborhood environment features are associated with long-term less decline in moderate-to-vigorous physical activity across the lifespan

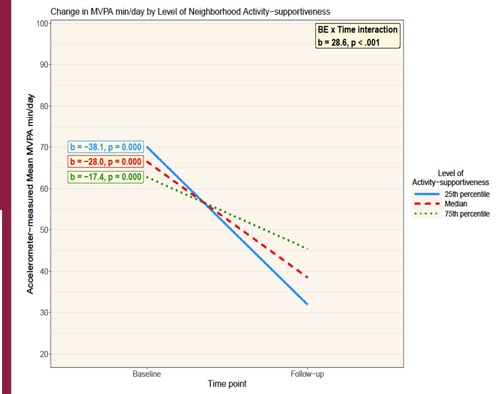
Table 1: Characteristics of study sample (N=1067)

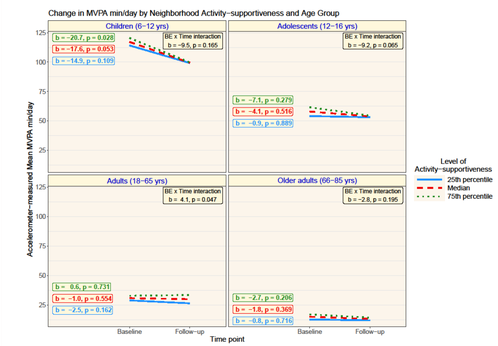
Study	Age at baseline	N with follow-up[a]	Follow-up period*	Age at follow-up[b]	% girls/ women	% college degree	% white/ non-hispanic	Baseline MVPA	Follow-up MVPA
NIK	6-11	262	6.1-9.8 yrs	16.1 (1.5)	51.0	60.6[c]	67.4	60.5 min/day	34.3 min/day
TEAN	12-16	163	5.1-9.0 yrs	20.5 (1.5)	52.4	66.1[c]	69.0	60.7 min/day	31.6 min/day
NQLS	18-65	430	9.9-15yrs	59.5 (9.9)	50.3	75.8	78.4	33.3 min/day	30.4 min/day
SNQLS	≥66	212	7.3-12 yrs	81.6 (4.9)	52.3	58.4	74.3	18.3 min/day	9.2 min/day

*SAFE study Final Sample (N=1,067), [b]Mean age (SD), [c]Reflect Parent's Education Status

Table 2: Model-predicted Mean MVPA min/day by BE Feature Level and Time with Tests of Interaction Terms in the Overall Sample

	Baseline		Follow-up		BE Feature x Time Interaction		
Level of BE Feature	Mean	95% CI	Mean	95% CI	b	95% CI	p
Overall activity-supportiveness							
25th percentile	70.1	(65.4, 74.9)	32.0	(27.6, 36.4)	28.65	(21.90, 35.39)	< 0.001
Median	66.5	(62.1, 71.0)	38.5	(34.5, 42.5)			
75th percentile	62.8	(58.0, 67.6)	45.4	(41.0, 49.8)			
Land use mix diversity							
25th percentile	72.1	(67.3, 76.9)	35.6	(31.1, 40.1)	10.24	(6.64, 13.83)	< 0.001
Median	68.5	(64.0, 72.9)	39.6	(35.6, 43.7)			
75th percentile	65.2	(60.4, 70.0)	43.2	(38.9, 47.6)			
Proximity to nearest transit stop							
25th percentile	71.6	(66.8, 76.4)	36.6	(32.1, 41.1)	6.08	(3.43, 8.72)	< 0.001
Median	68.1	(63.6, 72.6)	39.2	(35.1, 43.3)			
75th percentile	64.6	(59.7, 69.6)	41.8	(37.3, 46.3)			
Proximity to nearest park/recreation facility							
25th percentile	71.8	(66.8, 76.7)	34.9	(30.2, 39.6)	8.00	(4.89, 11.12)	< 0.001
Median	68.1	(63.6, 72.6)	39.2	(35.2, 43.3)			
75th percentile	64.4	(59.3, 69.6)	43.6	(38.9, 48.2)			
Aesthetics							
25th percentile	69.1	(64.4, 73.8)	36.7	(32.4, 41.1)	7.55	(2.04, 13.05)	0.007
Median	69.3	(64.8, 73.8)	40.6	(36.6, 44.7)			
75th percentile	69.4	(64.3, 74.6)	44.6	(39.8, 49.3)			
Street connectivity							
25th percentile-	69.2	(64.7, 73.7)	39.5	(35.5, 43.6)	4.08	(0.02, 8.15)	0.049
75th percentile	68.2	(62.9, 73.4)	42.6	(37.7, 47.4)			
Crime safety							
25th percentile	68.5	(63.9, 73.1)	37.9	(33.8, 42.1)	3.70	(-0.65, 8.05)	0.095
75th percentile-	70.7	(65.7, 75.6)	43.8	(39.2, 48.4)			
Pedestrian infrastructure							
25th percentile	71.8	(67.1, 76.4)	36.3	(32.1, 40.6)	13.42	(9.44, 17.39)	< 0.001
Median	67.9	(63.4, 72.4)	41.4	(37.3, 45.5)			
75th percentile	66.0	(61.3, 70.7)	43.9	(39.6, 48.3)			
Residential density							
25th percentile	70.2	(65.4, 75.0)	35.7	(31.3, 40.1)	0.16	(0.10, 0.22)	< 0.001
Median	68.9	(64.3, 73.6)	37.4	(33.2, 41.6)			
75th percentile	67.2	(62.6, 71.7)	39.7	(35.7, 43.8)			
Traffic/Pedestrian safety							
25th percentile	70.7	(66.1, 75.4)	37.1	(32.9, 41.3)	12.29	(6.90, 17.67)	< 0.001
Median	69.2	(64.7, 73.6)	39.6	(35.6, 43.7)			
75th percentile	67.6	(63.0, 72.2)	42.2	(37.9, 46.4)			





CONCLUSIONS

- MVPA declines from baseline to follow-up in all age groups
- 5 macro and 3 of 4 micro-scale baseline activity supportive-neighborhood environment features, and overall activity supportiveness are associated with less decline in MVPA across all age groups
- Those living in high-activity supportive neighborhoods have less decline in long-term MVPA compared to those in low-activity supportive neighborhoods

IMPLICATIONS OF FINDINGS

- Findings will augment the evidence base on associations of neighborhood environment with PA to better guide policy and practice around NCD prevention and control across the lifespan.

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